

Advanced Computer Architecture

High Performance Processors and Systems

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Name
Last Name
ID Number

Question 1 (20%)	
Question 2 (20%)	
Problem 1 (30%)	
Problem 2 (30%)	
Total (100%)	

To pass the midterm, a minimum score of 20% is requested by each “section” (section 1: problems, section 2: questions). The overall score, to consider the midterm as passed to access the project, has to be greater or equal to 50%

Question 1 (write your answer on the paper)

Define the main characteristics of a VLIW architecture. Explain which is the basic idea behind Trace Scheduling for VLIW and why is it necessary to get good performance from a VLIW?

Question 2 (write your answer on the paper)

Using the “Early Evaluation of PC” applied to branch instructions, how many stalls are needed to execute the following assembly code. Effectively support your answer

```
lw $1, 0($5)
beq $1, $0, label
```

Circle the bullet of the right answer

- **Answer 1:** 1
- **Answer 2:** 2
- **Answer 3:** 0
- **Answer 4:** 3

Answer

Answer 2 is the TRUE one. If a comparison register is a destination of immediately preceding load instruction 2 stall cycles are needed, the data to be stored in \$1 is available after the MEM stage (lw) and needed in the ID stage.

Problem 1

Assume that the following code has been executed on a CPU with TOMASULO with the following configuration:

- 2 RS (RS1, RS2) with 2 MUL unit and 10 CC of latency
- 2 RS (RS3, RS4) 2 ADDD/SUBD unit and 1 CC of latency
- 1 RS (RS5) 1 LDU and 2 CC of latency

	Instruction	ISSUE	Start EXE	WB
I1	LD F1, 0 (R2)	1	2	4
I2	MULTD F2, F1, F1	2	5	15
I3	ADDD F3, F1, F5	5	5	6
I4	MULTD F2, F3, F1	4	7	17
I5	SUBD F5, F1, F5	3	6	7

A. List all the possible conflicts in the code.

B. Is the proposed table correct? If it is not correct, please, write the right one.

Answer 1.A (write your answer on the paper)

RAW F1 I1-I2

RAW F1 I1-I3

RAW F1 I1-I4

RAW F1 I1-I5

RAW F3 I3-I4

WAW F2 I2-I4

WAR F5 I3-I5

Answer 1.B (write your answer here)

	Instruction	ISSUE	Start EXE	WB
I1	LD F1, 0 (R2)	1	2	4
I2	MULTD F2, F1, F1	2	5	15
I3	ADDD F3, F1, F5	3	5	6
I4	MULTD F2, F3, F1	4	7	17
I5	SUBD F5, F1, F5	5	6	7

Problem 2 (write your answer on the paper)

Describe (the answer has to be effectively supported) a 1-BHT and a 2-BHT able to execute the following assembly code (R0 is set to 1, R1 is set to 0)

```
LOOP:      LD      F1  0    R0
           ADDD    F2  F1   F1
           ADDI    R1  R1   1000
LOOP2:     MULTD   F2  F2   F1
           SUBI    R1  R1   1
           BNEZ    R1  LOOP2
           SUBI    R0  R0   1
           BNEZ    R0  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

We may compute all the miss predictions, considering all the possible initialization of the BHT, or we can see that with a R0 set to 1, after the first iteration SUBI will set R0 to 0, which means that LOOP, at the end, is not a LOOP, but a single iteration, therefore the provided code is not performing as expected in theory, where a nested loops scenario should benefit by a 2-bit BHT, while in this case, this is not happening.

LOOP executes 1 time
LOOP2 executes 1000 times

1-bit BHT, no collision

T,T: 1 + 1 : 2

T,NT: 1 + 2 : 3 W

NT,T: 0 + 1 : 1 B

NT,NT: 0 + 2 : 2

1-bit BHT, collision

T: 0 + 1 : 1

NT: 0 + 2 : 2

2-bit BHT, no collision

(Ts|Tw), (Ts|Tw): 1 + 1: 2

(Ts|Tw), NTw: 1 + 2: 3

(Ts|Tw), NTs: 1 + 3: 4 W

(NTs|NTw), (Ts|Tw): 0 + 1: 1 B

(NTs|NTw), NTw: 0 + 2: 2

(NTs|NTw), NTs: 0 + 3: 3

2-bit BHT, collision

(Ts|Tw): 1 + 1 : 2

NTw: 1 + 2: 3

NTs: 1 + 3: 4

B 1-bit and B 2-bit identical -> not in line w theory because external loop is only 1 iteration